Politecnico di Torino

FONDAZIONE LINKS
PASSION FOR INNOVATION

Multimodal Emotion Recognition Using Large Audio-Language Models

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PROJECT GOALS

1. Design a multimodal AI system to detect emotion

2. Promote human-centered and ethical AI systems

VALUE PROPOSITIONS

IMPROVED EMOTIONAL WELL-BEING

DATA-DRIVEN HEALTH INSIGHTS

ENHANCED PATIENT SUPPORT SYSTEMS

RESPONSIBLE & HUMAN-CENTERED AI

SUSTAINABLE DEVELOPMENT GOALS

PROBLEM STATEMENT

NOISE IN REAL-WORLD SPEECH

Existing systems fail when audio is noisy or text is ambiguous

FAILURE TO CAPTURE SUBTLE EMOTIONS

Single-modality models cannot capture complex emotional cues

DATASET SHIFT & GENERALIZATION ISSUES

Datasets differ widely, causing generalization issues

HYPOTHESES

NEED FOR EMOTION-SPECIFIC TRAINING

General-purpose audio-language models will perform poorly in recognizing emotions in speech without task-specific training.

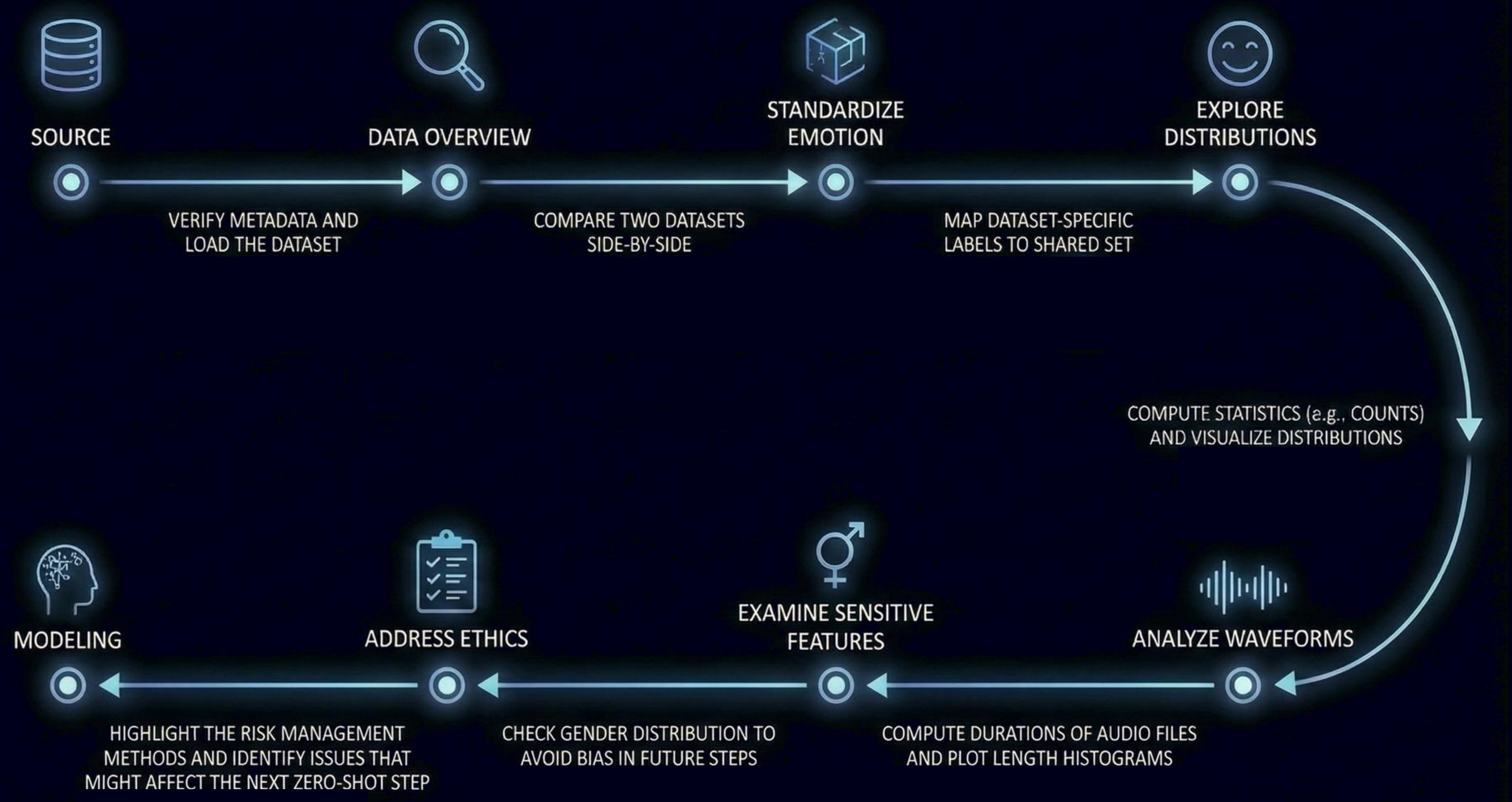
QA VS MULTIMODAL TRAINING

While transcripts contribute only marginal improvements because emotion cues are mostly acoustic, transitioning from QA-only training to emotion-focused multimodal training still results in better performance.

LORA FINE-TUNING BENEFITS

PEFT/LoRA fine-tuning on a balanced dataset (ESD) will significantly improve recognition of high-arousal emotions such as Angry and Happy.

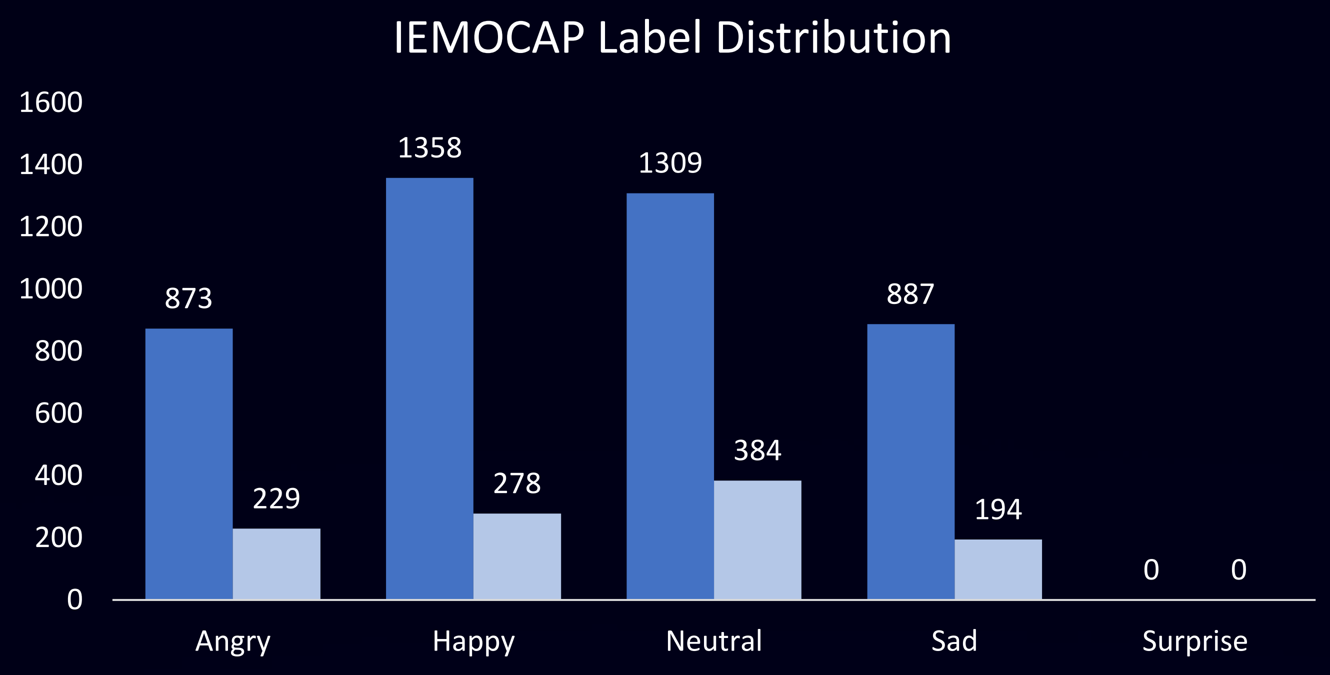
DATA EXPLORATION ROADMAP

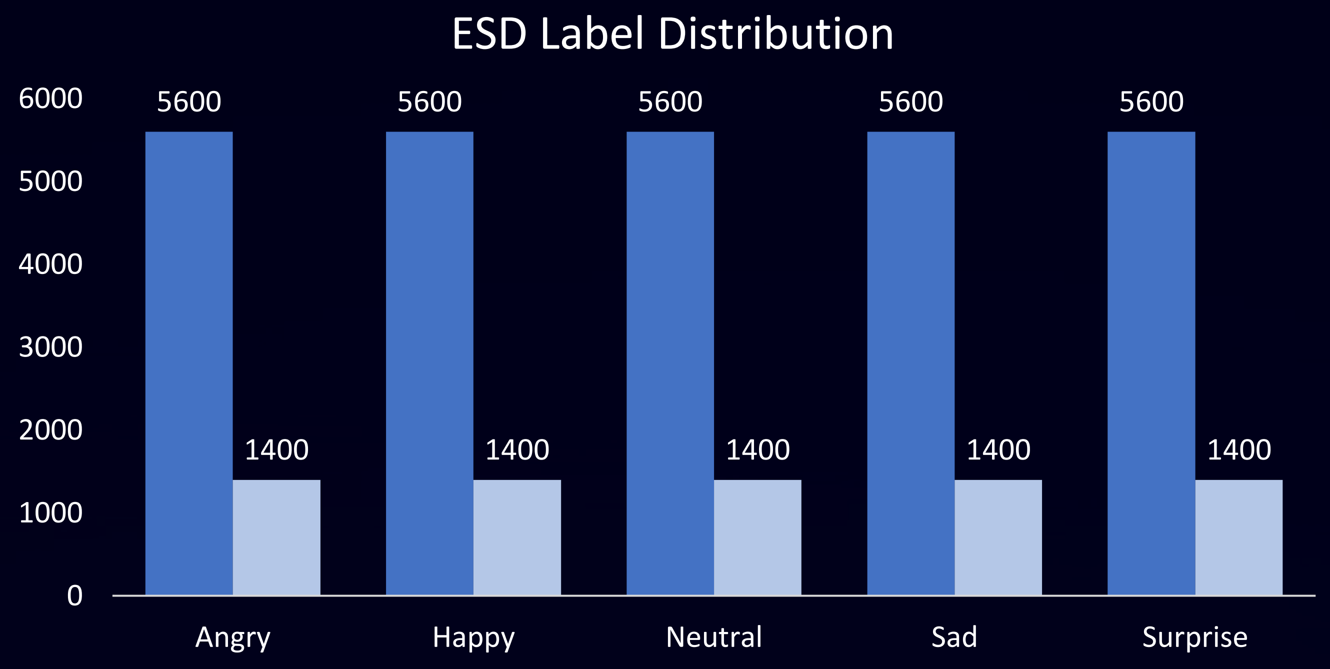


DATA SOURCE

	ESD	IEMOCAP
Language	English (This subset is used) + Chinese	English
Emotion Classes	5 (Angry, Happy, Sad, Neutral, Surprise)	4 (Angry, Happy/Excited, Sad, Neutral)
Class Balance	**Balanced**	**Imbalanced**
Speakers	20 speakers	10 actors
Recording Type	Read speech (controlled environment)	Natural dialogues
Audio Style	Clear and consistent	More expressive and varied
Metadata	**Limited** (no gender/age)	**Richer metadata** (speaker, gender)
Dataset Size	17500 samples (15750 Tr + 1750 Te)	5531 samples (4446 Tr + 1085 Te)
Use In Our Project	Good for **fine-tuning (PEFT)**	Good for **zero-shot testing**

EMOTION LABEL DISTRIBUTION: ESD VS IEMOCAP





IEMOCAP exhibits strong class imbalance

- Emotion frequencies vary significantly:
  - Neutral appears most often
  - Surprise does not exist in IEMOCAP
  - Angry and Sad are the least represented
- Realistic but imbalanced → more difficult for zero-shot generalization.

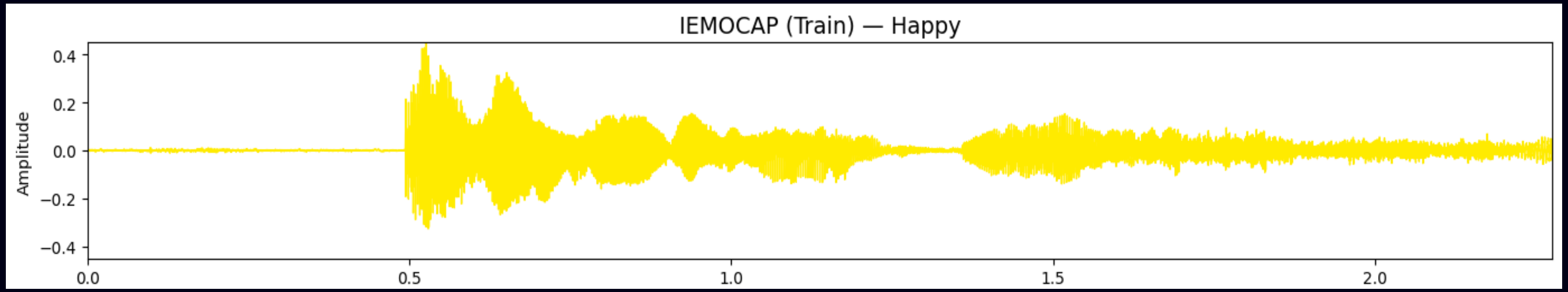
ESD shows completely balanced emotion distribution

- Perfectly balanced in train and test
- Designed dataset with controlled recordings
- Ideal for training and PEFT fine-tuning

WAVEFORM ACROSS EMOTIONS

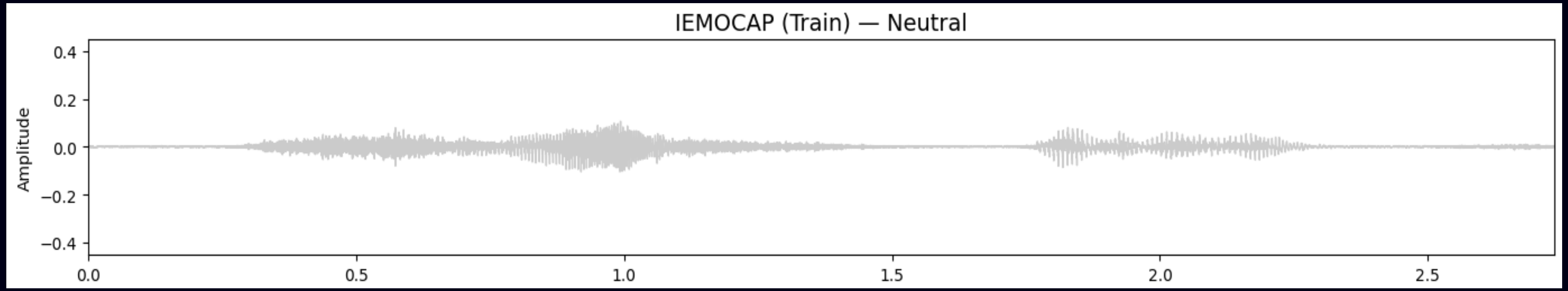
Happy

- Amplitude variations and frequent peaks



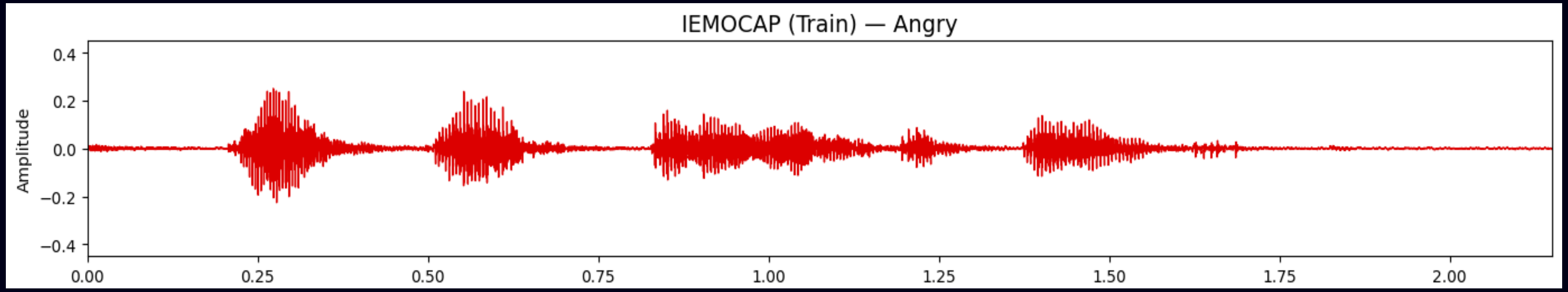
Neutral

- Low amplitude, flat contour, steady and calm



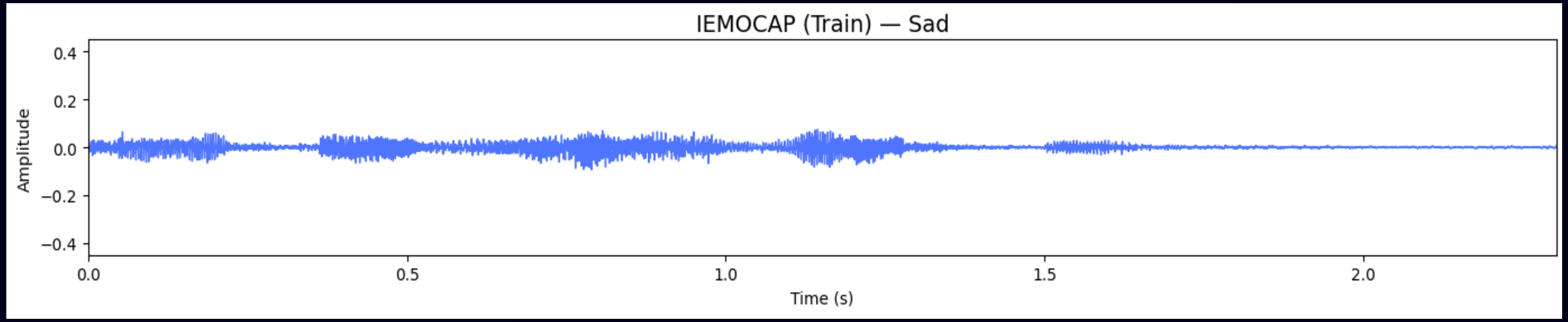
Angry

- Sharp, irregular spikes with high intensity



Sad

- Low intensity, smooth and longer contours



DATA PRIVACY

NO PERSONALLY IDENTIFIABLE INFORMATION

- Audio in both datasets contains only speech, no names or private identifiers
- No demographic fields beyond *speaker ID* and *emotion label*

STRICT NON-SHARING OF RAW AUDIO

- Raw .wav files remain local to the researchers and are never uploaded to GitHub
- Only preprocessed, non-reversible features are stored in the repository

METADATA IS USED FOR FAIRNESS EVALUATION

- Gender inferred from filenames solely for bias and fairness analysis, not for profiling
- No attempt is made to reconstruct or identify speakers.

ETHICAL HANDLING OF SENSITIVE ATTRIBUTES

- Sensitive analyses reported at group level, never at individual level
- Any demographic bias identified will inform model evaluation—not personalization

ZERO-SHOT EMOTION ANALYSIS APPROACHES

Voxtral Mini 3B

Unified Audio and Text Model
Handles both modalities in a single model.

Whisper-based Encoder
Uses Whisper Large v3 with 4x downsampling.

32k Token Context
Supports 30-40 min audio.

Designed for Transcription
Optimized for semantic audio understanding.

Qwen2 Audio 7B

Audio Enabled Qwen2 Model
7B parameters, multimodal capabilities.

Flexible Input
Accepts audio or audio and text directly.

Stronger Linguistic Reasoning
More capable than Voxtral, but slower with higher GPU cost.

Two Model Pipeline (ASR + Text LLM)

ASR

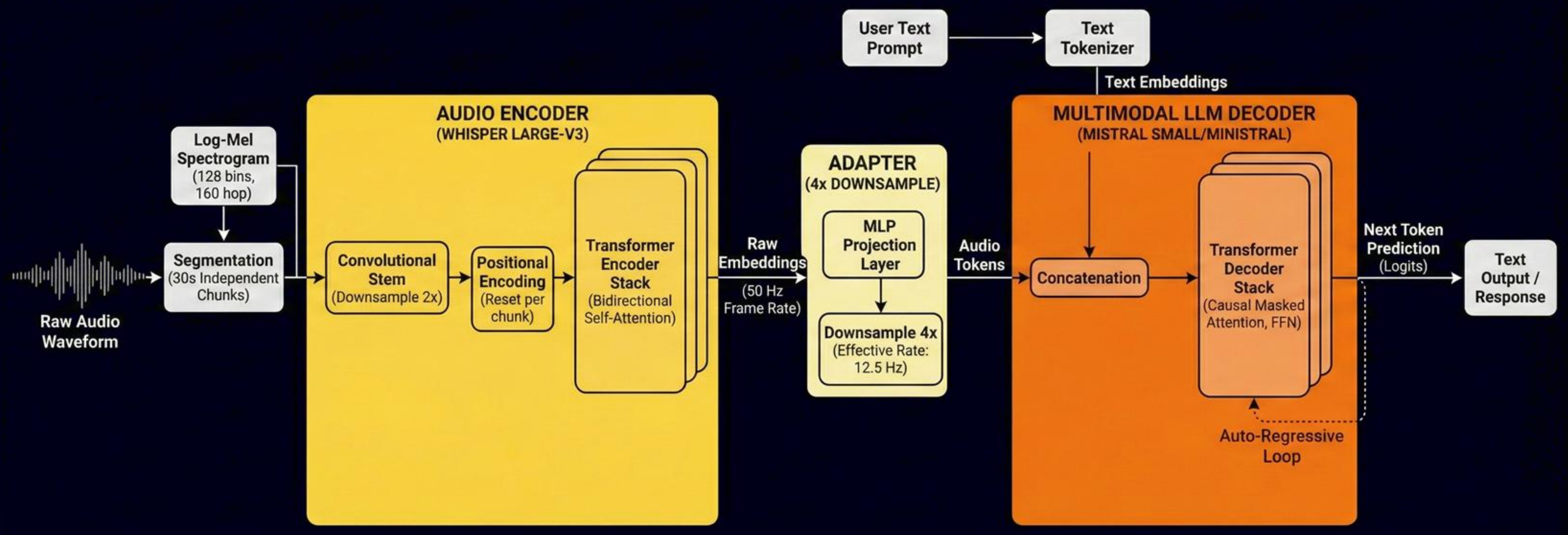
- Whisper Large V3 XLS R
- Stable transcripts, Flexible text reasoning
- Higher complexity, Error propagation from ASR

Text LLM

- Any text-only model (e.g., Qwen2.5 3B)
- Allows targeted fine-tuning
- Focus on small emotion datasets

WHY VOXTRAL-MINI-3B?

- One multimodal model removes ASR→LLM error chains, making the system simpler and more reliable.
- The 3B model fits Colab limits, while Qwen Audio 7B is heavier, slower, and harder to run on the full dataset.
- The model combines audio and text natively, so it naturally understands both prosody and meaning together.
- It supports long inputs (up to 32k tokens), allowing us to process full utterances without splitting the audio.



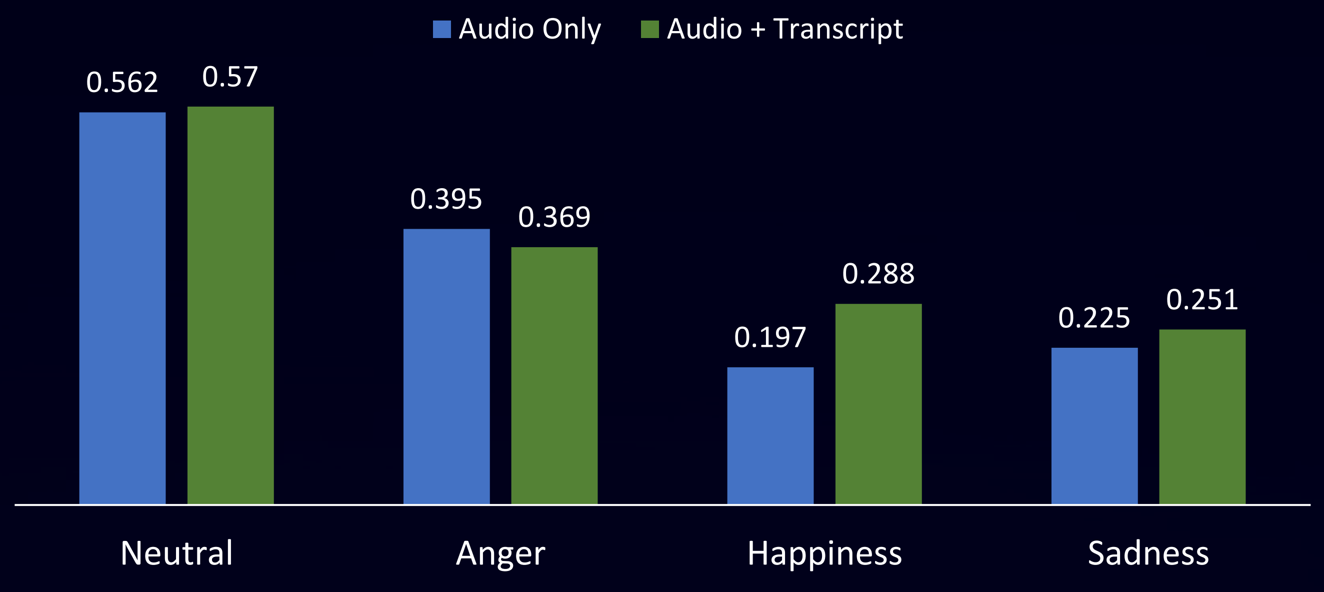
Yellow: Whisper Encoder, Light Yellow: Adapter, Orange: Mistral Decoder

ZERO SHOT PERFORMANCE ON IEMOCAP

	AUDIO ONLY	AUDIO + TRANSCRIPT
Accuracy	0.45	0.46
Balanced accuracy	0.37	0.38
Macro F1	0.34	0.37
Cohen's kappa	0.17	0.20
MCC	0.26	0.29

- Neutral Over Prediction Drives Metrics
- Limited Emotion Sensitivity in Zero Shot Mode
- Text Channel Gives Only Marginal Gains
- Low Macro-F1 Reflects Poor Class Separation

PER-EMOTION CLASS F1 SCORE

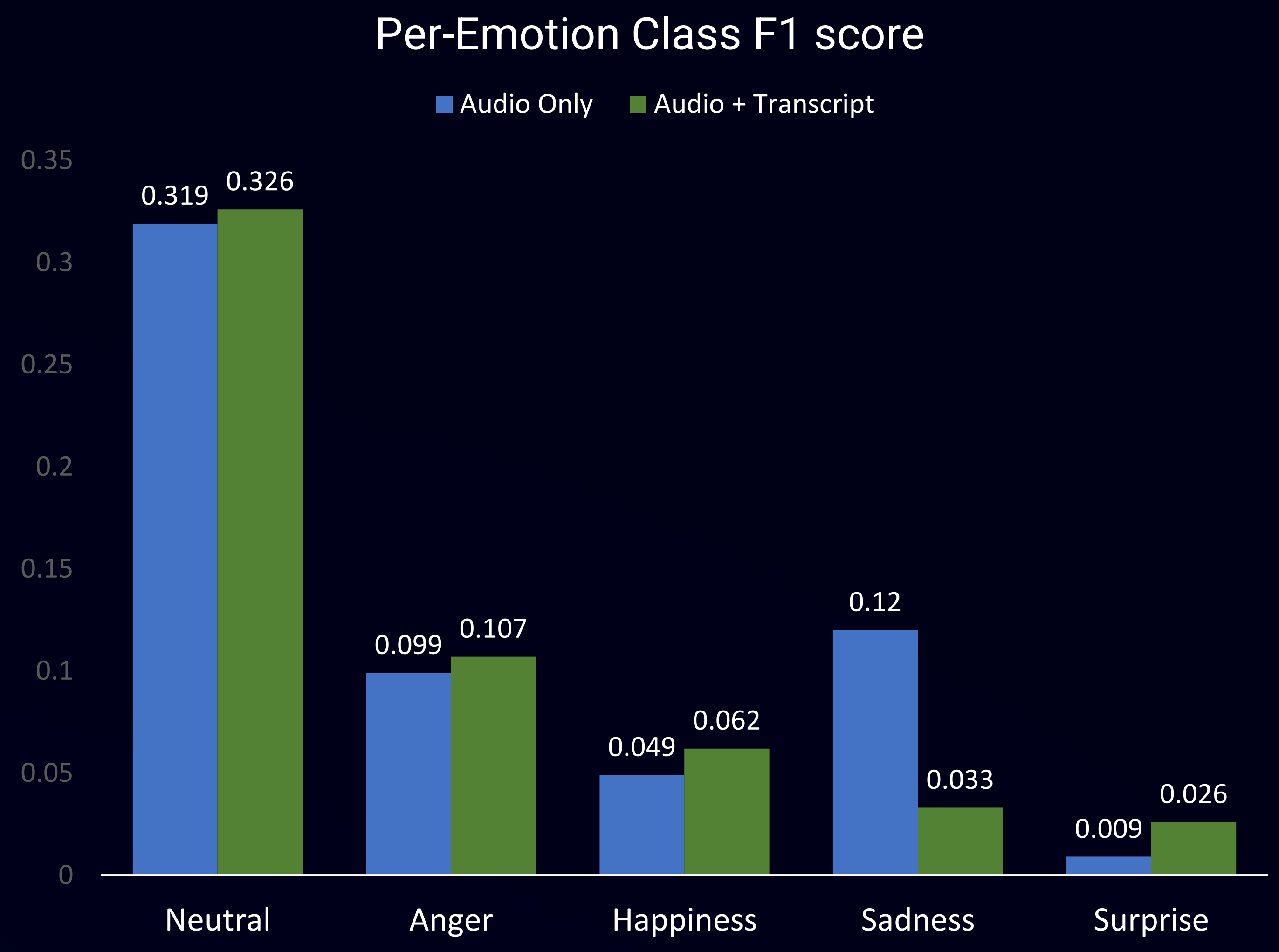


- Neutral stays almost the same with or without transcripts
- Anger drops slightly when transcripts are used
- Happiness improves the most with transcripts
- Sadness shows a small but consistent improvement

ZERO SHOT PERFORMANCE ON ESD

As seen in IEMOCAP, zero-shot results on ESD are near random, with only slight gains from adding transcripts. Neutral is the only emotion recognized reasonably well, while others remain poorly classified, confirming that effective emotion recognition requires fine-tuning.

	AUDIO ONLY	AUDIO + TRANSCRIPT
Accuracy	0.195	0.209
Balanced accuracy	0.195	0.209
Macro F1	0.113	0.128
Cohen's kappa	0.004	0.011
MCC	0.007	0.019



AGREEMENT AND DISAGREEMENT PATTERNS

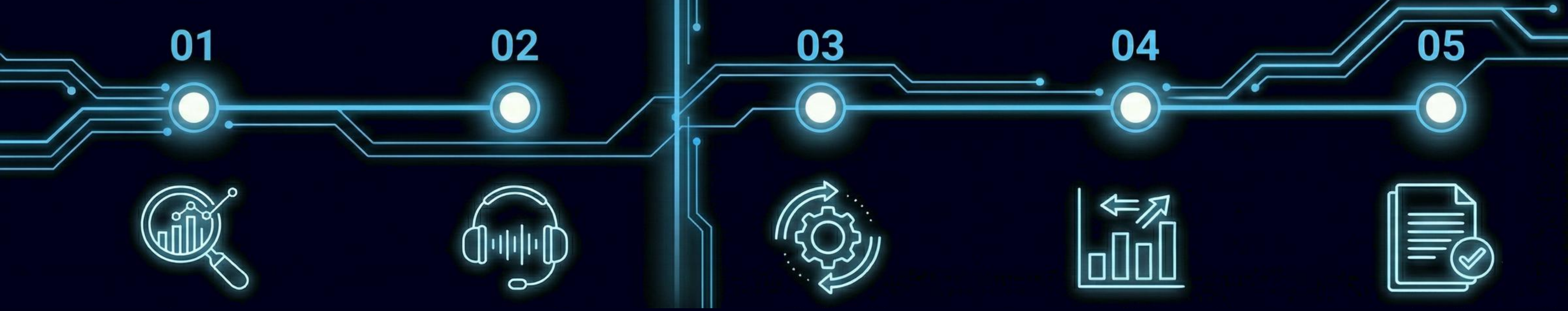
	ESD	IEMOCAP
Both Wrong	4110	568
Both Correct	980	472
A+T Correct	117	30
Audio only correct	43	15

- Errors often shared across modalities
- Both-correct cases are few
- Cross-modal fixes are rare
- Fusion does not correct individual errors

PROJECT PROGRESS AND NEXT STEPS

1. Defined objective, questions, and risk.
2. Explored ESD and IEMOCAP datasets.
3. Ran zero shot with Voxtral (audio/transcript).
4. Analyzed agreement/disagreement.

5. Apply PEFT fine tuning on Voxtral.
6. Re-evaluate hypotheses.
7. Compare zero shot vs PEFT.
8. Prepare updated analysis.



THANK YOU!